

No Genuine Copula in Japanese: a reanalysis of *da* as a predicative-postposition–adjunctive-conjunction complex plus a one-argument existential verb¹

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Abstract

This paper proposes an alternative analysis of the Japanese so-called copula *da* and its non-contracted form *dearu*. While *da/dearu* has traditionally been treated as a copula equivalent to English *to be*, we argue instead that it should be reanalysed as the complex consisting of the **predicative and conceptual-locative postposition =*ni*** and the **adjunctive conjunction -*te*** — surfacing as *=de* — combined with the **one-argument existential verb *aru***.

Under this analysis, *A=wa B=de aru* 'A is B' is semantically both an existence-asserting and an existence-domain-specifying sentence. Syntactically, it consists of the one-argument obligatory construction [*A=wa aru*] 'A exists' and the adjunct [*B=de*] 'and (A's existence is located) in the conceptual domain of B' or, more practically, 'and (it is) as B'. Schematically, this structure can be represented as the conjunctive proposition $\text{EXIST}(A) \wedge \text{IN}(\text{rEXIST}(A), B)$, distinct from the atomic $\text{BE}(A, B)$ encoded by *to be*. Although *B=de* is syntactically adjunctive, it is nevertheless non-deletable due to its pragmatic and informational indispensability: it contains the predicate-like postposition *=ni* and serves as the informational focus, whereas the syntactically obligatory *kare=wa aru* provides the presuppositional background.

This reanalysis appears to make possible a unified account of the structure, semantics and diachronic development of Japanese *da/dearu* constructions.

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1. Introduction — morphosyntactic and prosodic separability of *de* and *aru* in *dearu* and syntactic and prosodic constituency of *de* and the preceding NPs

In Japanese linguistics, *da/dearu* has traditionally been treated as a copula equivalent to *to be* in English, as in (1), where *dearu* can be replaced by *=da*, '=' indicating enclisis:

- (1) kare=wa gakusei dearu
 he(SUBJ)=TOP student(PRED) COP
 'He is a student.'

However, this analysis faces a major challenge: it cannot account for the **high degree of independence between *de* and *aru* in *dearu***. For instance, when the noun phrase (NP) *gakusei* 'student' is marked by the focus/additive enclitic particle *=mo* 'also' — one of the so-called *toritatesi* in Japanese linguistics — the sentence cannot take the expected form shown in (2):

- (2) *kare=wa gakusei =mo **dearu**
 he(SUBJ)=TOP student(PRED) also(ADD) COP
 'He is also a student.'

but must be as in (3):

- (3) kare=wa gakusei **de** =mo **aru**
 he(SUBJ)=TOP student(PRED) COP? also(ADD) COP?
 'He is also a student.'

The particle *=mo* is inserted between *de* and *aru*, as if behaving as an infix from the perspective that treats *dearu* as a copula — a situation comparable to the adverb *also* being inserted into the copula *is* in English, as in **he i- also -s a student*. Furthermore, in (4):

- (4) kare=wa gakusei **deari** kanozyo=wa syakaizin **=da**
 he(SUBJ)=TOP student(PRED) COP she(SUBJ)=TOP working adult(PRED) COP
 'He is a student, she is a working adult.'

the first *deari* may be reduced to *de* alone, without *ari* — formally, the conjunctive form, or *ren'yōkei*, of the verb *aru* — as in (5):

- (5) kare=wa gakusei **de** kanozyo=wa syakaizin **=da**
 he(SUBJ)=TOP student(PRED) COP? she(SUBJ)=TOP working adult(PRED) COP
 'He is a student, she is a working adult.'

Prosodically, too, the so-called copula *dearu* exhibits unexpected behaviour. If *dearu* were a single word, it would be expected to constitute either an independent prosodic word (ω) or an enclitic, so the utterance *kare=wa gakusei dearu* would be expected to be prosodised as either (6) or (7), where " | " marks a prosodic-word boundary, and prosodic words in Japanese generally correspond to traditional *bunsetsu*:

(6) | *kare=wa* | *gakusei=dearu* | — *dearu* as an enclitic (i.e. encliticised to *gakusei*)

(7) | *kare=wa* | *gakusei* | *dearu* | — *dearu* as an independent ω

In actual speech, nevertheless, the prosodisation is implemented as in (8):

(8) | *kare=wa* | *gakusei=de* | *aru* | — *de* as an enclitic and *aru* as an independent ω

Since prosodic-word boundaries are determined with reference to morphosyntactic information (Nespor & Vogel 2007), the fact that the utterance is prosodised as | *kare=wa* | *gakusei=de* | *aru* | strongly suggests the presence of a morphosyntactic boundary between *de* and *aru* — namely, that *de* and *aru* constitute distinct morphosyntactic elements. This prosodic separation is further supported by the fact that the discourse enclitic particle *=ne* — one of the so-called *syūzyōsi* in Japanese linguistics — can intervene between *de* and *aru*, as in (9), which would be impossible if *dearu* constituted a single prosodic word:

(9) | *kare=wa=ne* | *gakusei=de=ne* | *aru=ne* |

All of these facts demonstrate that ***de* and *aru* are morphosyntactically and prosodically separable**, whereas **the NP *gakusei* and *de* are always syntactically and prosodically inseparable**, as illustrated in (10):

(10) *Kare=wa* [*gakusei de*] [*aru*].

Kare=wa [*gakusei de*] *mo* [*aru*].

Kare=wa [*gakusei de*] [*ari*], *kanozyo=wa syakaizin=da*.

Kare=wa [*gakusei de*], *kanozyo=wa syakaizin=da*.

Kare=wa=ne [*gakusei de*]=*ne* [*aru*]=*ne*.

Hence, in *kare=wa gakusei dearu*, the sequence *gakusei=de* is assumed to constitute a syntactically and prosodically defined constituent, as Dalla Chiesa (2015) notes: "the NP-*de* element must always be the phrase (in the closest proximity to the verb)"², where the parenthetical remark is by the present author. This *gakusei de* is formally identical to a **postpositional phrase (PP) consisting of a NP combined with the so-called postposition =de**. In what follows, hence, *gakusei de* will be treated tentatively as a PP, that is, the PP *gakusei=de*. This separation between *=de* and *aru* is compatible with Matsuoka's observation that "the copula *da* consists of the postposition *de* and the verb *aru*" (2019: 142). The fact that *gakusei=de* and *aru* each form separate prosodic words can thus be naturally accounted for.

2. NP + *=de* as a conceptual-locative adjunct ConjP and *aru* as a one-argument existential verb

In Japanese, **PPs of the form [NP + *=de*] generally function as adjuncts**, as Hasegawa (1999: 26, translated from Japanese) states: "The case particle *=de* occurs not only for 'location of an action' but also for a variety of other elements such as 'cause' (*kaze=de yasumu* 'to be absent because of a cold'), 'means' (*genkin=de kau* 'to buy with cash'), 'time' (*mikka=de tukuru* 'to make in three days') and 'state' (*hadaka=de odoru* 'to dance naked'), and all of these fundamentally function only as adjuncts. There are no predicates that require these as arguments indispensable for their informational completeness. From this, one can generalise that *=de* marks adjuncts." Based on this generalisation, *kare=wa gakusei=de aru* can be syntactically analysed as consisting of the **obligatory structure [kare=wa aru]** and the **adjunctive structure [gakusei=de]**. Further justification for treating *gakusei=de* not as an argument but rather as an adjunct will be provided below in this section and in Section 8.

In *kare=wa aru*, the verb *aru* — or, more precisely, its stem *ar-* — can thus be analysed as a **ONE-ARGUMENT predicate**, taking only the NP *kare* as its argument. When the verb *aru* **functions as a one-argument predicate**, it means '**to exist**', as in *seigi=wa aru* 'justice exists'

² She argues that the ungrammaticality of **neko=de wagahai=ga aru* 'I am a cat' supports her claim that "the NP-*de* element must always be the phrase in the closest proximity to the verb". However, the ungrammaticality of this sentence does not arise from word order, but rather from the fact that *wagahai*, while functioning as a topic, is not marked by the particle *=wa*. Indeed, although fairly awkward, *neko=de wagahai=wa aru* is not completely ungrammatical, and expressions such as NP=*de watasi=wa ari-tai* 'as the referent denoted by the NP I want to exist' — e.g. *ryōsikitekina syakaizin=de watasi=wa ari-tai* 'as a rational member of society I want to exist' — sound much more natural.

— although in Contemporary Japanese this use often has a somewhat literary flavour, as in *ware omou, yueni ware ari* 'I think, therefore I am', and "**aru**"=*to iu koto=wa dō iu koto=ka=to iu toi* 'the problem of what it means "to exist"'. The sentence *kare=wa aru* can thus be interpreted as 'he exists', **asserting the existence** of *kare*. This meaning is reflected in the commonly used compound nouns *arikata* and *ariyō*, which signify 'mode/manner of existing, ideal way to exist' and 'state/appearance of existing', respectively. Both are derived from the stem *ari-* of the verb *aru* in this sense — namely, 'to exist' — combined with the nominal formatives *-kata* 'mode, manner, way' and *-yō* 'state, appearance', respectively. Further justification for analysing *aru* in this construction as 'to exist' — a meaning that, for pragmatic and informational reasons, is rarely brought to the speaker's or hearer's awareness in ordinary categorical or identificational contexts of the type 'A = B' or, more precisely, 'A ⊆ B' — will be presented in Section 6.

As for the PP *gakusei=de*, on the other hand, although **PPs headed by =de** can denote, as mentioned above with reference to Hasegawa (1999: 26), a variety of semantic roles such as location, cause, means, time or state, the most plausible interpretation in this construction is **locative** — namely, its primitive sense — though not denoting a physical, but a **conceptual, location**. Conceptual location here refers to the localisation or anchoring of an entity within a conceptual space structured by domains mutually defined through semiological opposition, founded on an analogy with physical space. Consequently, *gakusei=de* means 'in the **conceptual domain** of "student"' or, more practically, 'as a student'.

For instance, the construction in (11):

(11) *Kare=wa syakaizin=de aru: sigatu=ni gakusei=kara syakaizin=ni natta.*

'He is a working adult: he moved from being a student to being a working adult in April.'

'*He exists **in** a working adult: in April became **from** a student **to** a working adult. (lit.)'

indicates that the same entity denoted by *kare* has moved within the conceptual space **from** the domain *gakusei* 'student' **to** the distinct domain *syakaizin* 'working adult', thereby relocating its existence **in** the latter domain. The postpositions *=de* 'in' in *gakusei=de*, *=kara* 'from' in *gakusei=kara* and *=ni* 'to' in *syakaizin=ni* are plausibly understood as **metaphorical extensions of the physical location, source and goal markers**, as in the construction in (12):

(12) *Kare=wa Tōkyō=de hataraitte=iru: sigatu=ni Kyōto=kara Tōkyō=ni yatte=kita.*

'He is working **in** Tokyo: he came **to** Tokyo **from** Kyoto in April.'

Given that constructions encoding conceptual space systematically copy the constructions encoding physical space — that is, the linguistic isomorphism between physical space and conceptual space — it can be argued that it is not that speakers intentionally choose a spatial metaphor, but rather that the grammar provides no alternative formal resources, and consequently that Japanese grammar appears to formally force conceptual category domains to be treated in spatial terms.

Syntactically, the core phrasal structure found in *kare=wa gakusei=de aru* can be analysed tentatively as $[_{VP_1} [_{PP} \textit{gakusei=de}] [_{VP_0} \textit{kare ar-}]]$. In the lower VP $[_{VP_0} \textit{kare ar-}]$ 'he exists', the stem *ar-* of the existential verb *aru* and the NP *kare* constitute the predicate and its theme argument, respectively. In the higher VP $[_{VP_1} [_{PP} \textit{gakusei=de}] [_{VP_0} \textit{kare ar-}]]$ 'and his existence is located in the conceptual domain of "student"', on the other hand, the postposition *=de* — or, more precisely, **the underlying postposition =ni contained within the surface postposition–conjunction complex =de**, as will be discussed below — **behaves as if it were a two-argument predicate, and locates the lower VP $[_{VP_0} \textit{kare ar-}]$ 'he exists', construed as a reified eventuality 'his existence', within the conceptual domain denoted by the adjunctive NP *gakusei* 'student'**. Semantically, therefore, the construction expresses 'He exists, and **his existence is located in the conceptual domain of "student"**' or, more practically, 'He exists, and it is as a student'. Since the adjunct *gakusei=de* — containing the predicate-like *=ni* within *=de* — is introduced precisely for such a predication, **the most crucial logical** — though not syntactic — **predicate in this construction is** not the verbal stem *ar-* — which asserts *kare*'s existence in the lower VP $[_{VP_0} \textit{kare ar-}]$ — but rather **the predicate-like postposition =ni** contained within the postposition–conjunction complex *=de* — which, in the higher VP $[_{VP_1} [_{PP} \textit{gakusei=de}] [_{VP_0} \textit{kare ar-}]]$, locates *kare*'s existence denoted by the lower VP $[_{VP_0} \textit{kare ar-}]$ in the conceptual domain referred to by the NP $[_{NP} \textit{gakusei}]$.

Schematically, the more abstract construction *A=wa B=de aru* — in which *A=wa aru* 'A exists', even without *B=de* 'and (A's existence is located) in the conceptual domain of B', constitutes an atomic proposition, since *B=de* is assumed here to be adjunctive — can be represented as **E(A) $\wedge$ IN(rE(A), B)**. Here the first proposition E(A) 'A exists' — where E abbreviates 'exist' — denotes **the assertion of existence**, whereas the other proposition IN(rE(A), B) 'A's existence is located in the conceptual domain of B' represents **the specification of the domain of existence**. The function r in rE(A) — which abbreviates 'reified' — maps the proposition E(A) 'A exists' to the corresponding reified eventuality rE(A), that is, an eventive entity representing 'A's

existence'. This analysis requires a three-tier mapping:

- (i) from the proposition $E(A)$ 'A exists';
- (ii) to the actual eventuality that makes $E(A)$ true;
- (iii) to the reified eventuality $rE(A)$ corresponding to 'A's existence', which functions as an entity capable of serving as x in the relational predication $IN(x, B)$, headed by the predicate-like $=ni$ within the postposition–conjunction complex $=de$.

The structure $E(A) \wedge IN(rE(A), B)$ thus constitutes a conjunctive proposition. Note that, since $IN(rE(A), B) \Rightarrow E(A)$, it follows that $E(A) \wedge IN(rE(A), B) \equiv IN(rE(A), B)$; hence **$E(A) \wedge IN(rE(A), B)$ is logically redundant**, although it is still assumed to be **indispensable for a faithful formalisation of the syntactic–semantic structure**.

Let us return to the adjunctivity of the PP $B=de$ in $A=wa B=de aru$. As will be discussed in Section 7, the contemporary postposition $=de$ originated historically from the sequence $=ni-te$, where $=ni$ is assumed to have functioned as a locative postposition and $-te$ as a conjunctive suffix. The development was probably driven by the deletion of $-i-$ and by postnasal voicing — typical of the Yamato lexical stratum — yielding the sequence $=ni-te \rightarrow =n-te \rightarrow =n-de \rightarrow =de$. Even today, in PPs of the form $[NP + =de]$, one can observe stylistic variation between the postposition $=de$ and the postposition–suffix sequence $=ni-te$, as in $Tōkyō=de / Tōkyō=ni-te$ 'in Tokyo, at Tokyo', though the latter is a more formal and literal form. Given this diachronic development and the synchronic variation, it is plausible to assume that the underlying form of $B=de$ is still $B=ni-te$, where, strictly speaking, **$B=ni$ denotes conceptual location and the conjunctive suffix $-te$ adjoins $B=ni$, not as an argument but as an adjunct**, to $A=wa aru$ — or more precisely, **to the VP node $[A ar-]$ within $A=wa aru$** .

An NP marked by $=ni$ functioning as an argument is required by the predicate as an element semantically indispensable for completing the predicate's meaning. The link between the predicate and such an argumental $NP=ni$ is determined by the argument structure inherent in the predicate, and is therefore intrinsic and necessary to it. By contrast, an underlying $NP=ni$ functioning as an adjunct is optional for the predicate or for some node in the predicate phrase, and the link between the adjunct $NP=ni$ and the predicate or such a node is extrinsic and contingent to the predicate. In Contemporary Japanese, this extrinsic and contingent relation

needs to be made explicit in the linguistic form, and the device used for this purpose is the conjunctive suffix *-te*. It may be assumed, from a cognitive perspective, that the *conjunctivity* of *-te* has been metaphorically extended — via the schema of *junctivity* — into *adjunctivity*, as schematised in (13):



Having thus developed into an adjunctive suffix, *-te* adjoins *NP=ni* to the predicate or to some node in the predicate phrase as an adjunct rather than as an argument. The surface realisation of this underlying *NP=ni-te* is *NP=de*. Thus, the use of *=de* as an adjunct marker in Contemporary Japanese can be accounted for by the fact that its underlying form *=ni-te* contains the conjunctive suffix *-te*, which functions adjunctively in this context. Consequently, *B=ni-te*, surfacing as *B=de*, can be analysed not as a PP but as a conjunction phrase (ConjP) headed by the conjunction *-te* with the complement PP *B=ni*, as illustrated in (14):

(14) [ConjP [PP [NP *B*] [P *=ni*]] [Conj *-te*]]

In the construction *A=wa B=de aru*, therefore, it seems accurate to characterise *=de* as a **complex consisting of the predicative and conceptual-locative postposition *=ni* and the adjunctive conjunction *-te***.

The remaining problem is to determine what element in the linguistic form *A=wa B=de aru* formally corresponds to the operator $\wedge$, which expresses logical conjunction in the proposition $E(A) \wedge \text{IN}(\text{rE}(A), B)$. The phrasal structure corresponding to the schematic representation $E(A) \wedge \text{IN}(\text{rE}(A), B)$ is assumed to be as in (15) in line with (14):

(15) [VP₁ [ConjP *B=ni-te*] [VP₀ *A ar-*]]

where the lower VP [VP₀ *A ar-*] corresponds to the proposition $E(A)$, and the higher VP [VP₁ [ConjP *B=ni-te*] [VP₀ *A ar-*]] — formed by adjoining the adjunct PP [PP *B=ni*] to the lower VP [VP₀ *A ar-*] via the adjunctive conjunction *-te* — corresponds to $\text{IN}(\text{rE}(A), B)$. The semantic composition corresponding to the structure in (15) is given in (16):

(16) [NP <i>A</i>]:	A
[V <i>ar</i> -]:	E()
[VP ₀ <i>A ar</i> -]:	E(A)
[NP <i>B</i>]:	B
[P <i>=ni</i>]:	IN(r(),)
[Conj <i>-te</i>]:	Λ
[PP <i>B=ni</i>]:	IN(r(), B)
[ConjP <i>B=ni-te</i>]:	Λ IN(r(), B)
[VP ₁ <i>B=ni-te A ar</i> -]:	E(A) ∧ IN(rE(A), B)

As stated above, E(A) is the proposition 'A exists'; rE(A) is the actual eventuality that makes this proposition true and, when reified, yields the eventive entity 'A's existence'; IN(rE(A), B) is the other proposition 'A's existence is located in the conceptual domain of B'; and what corresponds to the logical conjunction operator $\wedge$, which combines these two propositions, is the adjunctive conjunction *-te*. Accordingly, the conjunction *-te* both syntactically adjoins the PP [PP *B=ni*] to the VP [VP₀ *A ar*-] as an adjunct, and, logically, functions as the conjunctive operator $\wedge$, introducing the predicate IN(r(),), denoted by the predicate-like postposition *=ni*. Thus, the lower VP [VP₀ *A ar*-] already constitutes an independent proposition E(A) and, by merging this lower VP with the adjunctive ConjP [ConjP *B=ni-te*], which expresses $\wedge$ IN(r(), B), the higher VP [VP₁ *B=ni-te A ar*-] is formed, representing the additional proposition $\wedge$ IN (rE(A), B).

In opposition to this *A=wa B=de aru*, the semantically related and syntactically similar construction *A=wa B=ni aru/iru* 'A is located in B' — in which *A=wa aru/iru* 'A is located', without *B=ni* 'in B', cannot constitute a proposition, since *B=ni* is argumentative, as discussed in Section 4 — can be represented as **IN(A, B)**, which denotes the **specification of the domain of location**. IN(A, B) thus constitutes an **atomic proposition**. The compositional difference between the conjunctive E(A) $\wedge$ IN(rE(A), B) and the atomic IN(A, B) is reflected in their default negation patterns, such as the partial negation *A=wa B=de=wa nai* in the former vs. the total negation *A=wa B=ni nai/i-nai* in the latter — namely, E(A) $\wedge$ $\neg$ IN(rE(A), B) vs. $\neg$ IN(A, B) — which will be discussed in Section 8.

Thus, instead of the traditional **copula-based** analysis in (17) = (1), representable as BE(A, B):

constructions *kare=wa hataraitte=iru* and *kare=wa aru*, respectively — to the corresponding reified eventualities $r\text{WORK}(kare)$ and $r\text{EXIST}(kare)$. These reified eventualities are then located by the underlying postposition *=ni* — amalgamated with *-te* to surface as *=de* — within the domains denoted by the NPs *Tōkyō* and *gakusei*, yielding additional propositions $\text{IN}(r\text{WORK}(kare), Tōkyō)$ and $\text{IN}(r\text{EXIST}(kare), gakusei)$. The underlying conjunction *-te* adjoins these eventuality-domain-specifying propositions to the base propositions $\text{WORK}(kare)$ and $\text{EXIST}(kare)$, yielding the conjunctive propositions $\text{WORK}(kare) \wedge \text{IN}(r\text{WORK}(kare), Tōkyō)$ and $\text{EXIST}(kare) \wedge \text{IN}(r\text{EXIST}(kare), gakusei)$. Accordingly, it can be assumed that **the Japanese postposition–conjunction complex =de** does not function as a simple locative postposition but rather as what can be called an **eventuality-domain specifier**: $\wedge (r(), \text{NP})$.

4. The differences between *sore wa gakkō=DE ARu* and *sore wa gakkō=NI ARu*

In contrast to the construction *sore=wa gakkō=de aru* in (21), hypothesised as **existence-asserting and existence-domain-specifying** in Section 2:

- (21) *sore=wa* *gakkō=de* **ar-u**
 it(SUBJ)=TOP school=CONCEP-LOC-CONJ(ADJUNCT) **exist-PRES**
 'It exists, and its existence is located in the conceptual domain of "school".'

in genuine locative — namely, **location-specifying-only** — constructions as in (22) and (23), where **PHYSIC-LOC** abbreviates 'physical-locative':

- (22) *sore=wa* *gakkō=ni* **ar-u**
 it(SUBJ)=TOP school=PHYSIC-LOC(ARGUMENT) **be located-PRES**
 'It is located in the physical domain of "school".'
- (23) *kare=wa* *gakkō=ni* **i-ru**
 he(SUBJ)=TOP school=PHYSIC-LOC(ARGUMENT) **be located-PRES**
 'He is located in the physical domain of "school".'

the **physical-locative argument is obligatory**, as mentioned in Section 2, and **always marked by the postposition =ni**, rather than *=de*. Thus, the verb *aru* — more precisely, its stem *ar-* — when used as a **LOCATIVE verb**, is a **TWO-ARGUMENT predicate with the argument structure [physical domain, theme]**, the location being denoted by the postpositional phrase $[\text{NP} + =ni]$, as in *sore=wa gakkō=ni aru*. When the theme is an animate entity, the verb *iru* with the stem *i-* is used instead as in (23), with the **argument PP** $[\text{NP} + =ni]$ **unchanged**, as in *kare=wa*

gakkō=ni iru 'he is in the school'. By contrast, the sequence of words in (24), in which the argumentative PP [NP + =*ni*] in (23) has been replaced by the ConjP [NP + =*de*]:

- (24) *kare=wa gakkō=de i-ru*
 he(SUBJ)=TOP school=LOC-CONJ(ADJUNCT) ?-PRES
 ' ? '

is semantically uninterpretable — even for native speakers — and ungrammatical, indicating that the ConjP [NP + =*de*] is adjunctive in this construction, which lacks an indispensable argument headed by =*ni*.

Given the general adjunctivity of the ConjP [NP + =*de*] and its specific adjunctive function in purely locative constructions such as (24), *kare=wa gakusei=de aru* may reasonably be regarded SYNTACTICALLY — though NOT PRAGMATICALLY OR INFORMATIONALLY, as will be discussed in Section 5 — as a **structurally self-contained, existence-asserting construction** *kare=wa aru* 'he exists', **accompanied by the conceptual-locative adjunct** *gakusei=de* 'and (his existence is located) in the conceptual domain of "student"', **which conjunctively specifies the domain of existence**.

Thus, in *A=wa B=de aru*, the constituent *B* denotes a conceptual domain of location, whereas in *A=wa B=ni aru/iru*, *B* denotes a physical domain of location. This difference surfaces formally when *B* is replaced by a pro-form. As shown in (25) and (26) below, in the former construction *B* is substituted by the pro-form *sō*, which expresses such notions as property, attribute, characteristic or manner and is glossable as 'so', whereas in the latter construction *B* is substituted by the locative demonstrative pronoun *soko* 'there':

- (25) *A=wa B=de aru.* → *A=wa sō=de aru.* (conceptual domain)

- (26) *A=wa B=ni aru/iru* → *A=wa soko=ni aru/iru.* (physical domain)

Although both constructions are assumed to involve a proposition of the form IN(*x*, *y*), the formal opposition between *sō* and *soko* reflects whether the second term *y* denotes a conceptual or, alternatively, a physical domain. Note that even when the postposition-conjunction complex =*de* — namely, the eventuality-domain specifier — is used, if *B* in *B=de* denotes a physical domain, it is substituted by *soko*, as in *kare=wa gakkō=de / Tōkyō=de hataraite=iru* 'he is working in a/the school / in Tokyo' → *kare=wa soko=de hataraite=iru* 'he is working there'.

5. The syntactic optionality and pragmatic necessity of $B=de$ in $A=wa\ B=de\ aru$

Although $B=de$ — underlyingly $B=ni-te$ — in $A=wa\ B=de\ aru$ is assumed to be **syntactically an adjunct**, it **cannot be deleted**. This is because, in this construction, **the most crucial logical** — though not syntactic — **predicate is** not the verbal stem $ar-$ that asserts A 's existence, but rather **the predicate-like postposition $=ni$ within the postposition–conjunction complex $=de$, which locates A 's existence in the conceptual domain referred to by B** . It is also because the sentence, while structurally consisting of the obligatory $A=wa\ aru$ 'A exists' and the adjunctive $B=de$ 'and (A's existence is located) in the conceptual domain of B' or 'and (it is) as B', functions as a categorical or identificational sentence in which **the focus falls on the adjunct $B=de$** .

Therefore, the **non-deletability of $B=de$** is assumed to **arise from its pragmatic and informational indispensability** rather than from syntactic necessity. The distinction can be represented as in the following table (27):

(27)

Level	Structure	Function
Syntax	$[_{VP}\ A\ ar-] + [_{ConjP}\ B=ni-te]$	One-argument VP + adjunctive ConjP
Semantics	'A exists, and A's existence is located in the conceptual domain B'	Existence assertion + domain specification $E(A) \wedge IN(rE(A), B)$
Information Structure	$[A=wa\ aru] + [B=de]$	Given + New (default)
Pragmatics	Focus on $B=de$ (default)	Pragmatic obligatoriness and non-deletability of $B=de$ due to focus

6. Shift of focus and foregrounding of the lexical meaning 'to exist' of the verb aru

In the construction $watasi=wa\ ry\osikitekina\ syakaizin=de\ aru$ in (28):

- (28) $watasi=wa$ $ry\osikitekina\ syakaizin=de$ **ar-u**
 I(SUBJ)=TOP rational member of society=CONCEP-LOC-CONJ(ADJUNCT) **exist-PRES**
 'I EXIST as a rational member of society.'

which can be schematically represented as $E(A) \wedge IN(rE(A), B)$, the existence-asserting component $E(A)$ — that is, the assertion of the lexical meaning 'to exist' encoded in the verbal stem $ar-$ — is backgrounded as given information, whereas the domain-specifying component $IN(rE(A), B)$ is foregrounded as the focus of the utterance. As a result, the meaning 'to exist' is rarely brought to the speaker's or hearer's awareness in ordinary categorical or identificational

contexts of the type 'A $\subseteq$ B'. By contrast, in example (29), which takes the form **A WANT [E(A) $\wedge$ IN(rE(A), B)]**, the situation is fairly different:

- (29) watasi=wa ryōsikitekina syakaizin=de **ari-** -tai-0
I(SUBJ)=TOP rational member of society=CONCEP-LOC-CONJ(ADJUNCT) **exist** want-PRES
'I want to EXIST as a rational member of society.'

Here, what is at issue is 'one's mode of existence', that is, an orientation toward a particular manner of being. In such a context, **E(A) is no longer backgrounded but instead becomes part of the focus as well as IN (rE(A), B), and the lexical meaning 'to exist' encoded in *ar-* emerges as perceptually salient.** This pragmatic shift of focus, which foregrounds the existential meaning of *ar-*, supports empirically the validity of analysing **the semantic structure of *A=wa B=de aru* not as BE(A, B) but as EXIST(A) $\wedge$ IN(rEXIST(A), B).**

It should be noted that the English translation *I want to BE a rational member of society*, representable as **A WANT [BECOME(A, B)]** — based on *I AM a rational member of society*, which is representable as **BE(A, B)** — is a complete mistranslation of the Japanese construction *watasi=wa ryōsikitekina syakaizin=de ARI-tai* in (29). This in itself clearly demonstrates that the Japanese *=da / =de aru* is not equivalent to the English *to be*, nor does it function as a copula in the same way. The difference between them can be represented as in (30):

- | | |
|---|--------------------------------------|
| (30) <i>Watasi</i> =wa ryōsikitekina syakaizin= de aru : | EXIST(A) ∧ IN(rE(A), B) |
| <i>I am</i> a rational member of society: | BE(A, B) |
| <i>Watasi</i> =wa ryōsikitekina syakaizin= de ari-tai : | A WANT [EXIST(A) ∧ IN(rEXIST(A), B)] |
| <i>I want to be</i> a rational member of society: | A WANT [BECOME(A, B)] |

What corresponds in Japanese to the English *I want to be a rational member of society* in (26) is the sentence in (31):

- (31) *Watasi=wa ryōsikitekina syakaizin=ni nari-tai*: A WANT [BECOME(A, B)]

which can be represented as shown in (32), containing the two-argument state-change verb *naru* 'to become', with *=ni* marking the argument GOAL:

- (32) watasi=wa ryōsikitekina syakaizin=ni nari- -tai-0
I(SUBJ)=TOP rational member of society=GOAL(ARGUMENT) become want-PRES
'I want to BE a rational member of society.'

7. From Old Japanese to Contemporary Japanese forms

In Old Japanese (OJ, 700–800), the modern formally distinct constructions $A=wa\ B=de\ aru$ 'A exists (and it is) as B' and $A=wa\ B=ni\ aru/iru$ 'A is (located) in B' would have corresponded to the seemingly identical construction $A\ B=ni\ ari$. "Dative *ni* is the general oblique case, marking both argument and non-argument oblique nominals" (Frellesvig 2010: 126), unlike in Contemporary Japanese, where $=ni$ marks arguments, while $=de$ — which originated from the postposition–suffix sequence $=nite$ — marks adjuncts, in accordance with the diachronic description "OJ *ni* and *nite* became more specialized with *ni* being used more for arguments and *nite* more for adjuncts; from mid to late EMJ *nite* acquired the variant *de* which is still in use in contemporary NJ" (*ibid.*: 243), where EMJ and NJ abbreviate Early Middle Japanese (800–1200) and 'new Japanese' — namely, Modern Japanese, 1600– — respectively.

Thus, the former $A\ B=ni\ ari$ 'A exists (and it is) as B' can be analysed as an existence-asserting and existence-domain-specifying construction consisting of the obligatory construction $A\ ari$ 'A exists' — with the verb *ari* being used as **one-argument existential** predicate — and the conceptual-locative adjunct $B=ni$ 'and in the conceptual domain referred to by B', as shown in (33). The latter $A\ B=ni\ ari$ 'A is (located) in B', on the other hand, can be analysed as a location-specifying -only construction — in which *ari* is employed as a **two-argument locative** predicate — and $B=ni$ 'in B' constitutes the physical-locative argument, as indicated in (34):

- (33) A B=**ni** **ari-0**
 A(SUBJ)=TOP B=CONCEP-LOC(ADJUNCT) **exist-PRES**
 'A exists, its existence is located in the conceptual domain B.'

- (34) A B=**ni** **ari-0**
 A(SUBJ)=TOP B=PHYSIC-LOC(ARGUMENT) **be located-PRES**
 'A is located in the physical domain B.'

The Old Japanese constructions in (33) and (34) appear superficially identical, but when the element *B* is substituted by a pro-form, they diverge as in (35) and (36):

- (35) $A\ B=ni\ ari. \rightarrow A\ sa=ni\ ari.$ (conceptual domain)

- (36) $A\ B=ni\ ari. \rightarrow A\ soko=ni\ ari.$ (physical domain)

Consequently, although both constructions is assumed to involve a proposition of the form $IN(x, y)$, the pro-form opposition between *sa* and *soko* reflects whether the second term *y* refers to a

These superficially identical constructions in (33) and (34), however, diverged diachronically into the formally distinct modern constructions in (37) and (38):

- As already stated in Section 4, in Contemporary Japanese as well, the element *B* undergoes different pro-form substitutions, as in (39) and (40) = (25) and (26):

- In (38), the locative verbs *aru* and *iru*, both semantically corresponding to the Old Japanese *ari* — though the latter originally derived from another verb *wiru* 'to sit; to be seated' — are used for inanimate and animate entities, respectively. The historical development from (33) to (37) is as follows. In Old Japanese, there existed "analytic forms consisting of an infinitive (*ni* ...) followed by *ar-*" (Frellesvig 2010: 95) for the so-called copula — that is, *ni ar-*. In the subsequent Early Middle Japanese (EMJ, 800–1200) period, this analytic form "*ni ar-* had the variant *nite ar-*" (*ibid.*: 235). "In the course of EMJ" (*ibid.*: 235), or more specifically "from the end of the period" (*ibid.*: 233), "*nite ar-* became *de ar-*" (*ibid.*: 235). As illustrated in "Table 12.8. *Development of the copula and adjectival-copula paradigms*" (*ibid.*: 344), from the end of EMJ to early Late Middle Japanese (early LMJ, Kamakura period 1185–1333), the three forms *ni ar-*, *nite ar-* and *de ar-* coexisted, although "already from late EMJ" (Insei period, 1086–1185), "*ni* was getting more restricted" and "*de* was used far more widely" than *nite*. The same table (*ibid.*: 344) shows that only the newest *de ar-* continued to be used from late LMJ (Muromachi period 1333–1573) into Contemporary Japanese.

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(41) $A B=ni\ ari \rightarrow A B=ni-te\ ari \rightarrow A=wa\ B=de\ aru$ — $E(A) \wedge IN(rE(A), B)$

(42) $A B=ni\ ari \rightarrow A B=ni\ ari \rightarrow A=wa\ B=ni\ aru/iru$ — $IN(A, B)$

may plausibly reflect the diachronic split between *=ni*, used for arguments, and *=ni-te* — later, *=de* — used for adjuncts, where the conjunctive suffix *-te* adjoins the PP [NP + *=ni*] as an adjunct, rather than as an argument, to another node, as discussed in Section 2. This suggests a shift from a linguistic stage in which *=ni* marked both arguments and adjuncts, without the help of the conjunction *-te*, to one in which *=ni* and *=ni-te/=de* became specialised as argumental and adjunctive markers, respectively. Thus, the historical split of the Old Japanese *=ni ari* between the Modern Japanese *=de aru* and *=ni aru/iru* may not reflect a reanalysis of a single construction, but rather the formal divergence of two distinct semantic structures — 'existence and its location in a conceptual domain' vs. 'location in a physical domain' — that happened to share the same surface form *A B=ni ari* in Old Japanese — a possibility that merits further historical investigation.

8. Default partial negation and the scope of negation marked by the particle *=wa*

The **default negation pattern of the existence-asserting and existence-domain-specifying** sentence in (43) = (18):

(43) $kare=wa$ $gakusei=de$ $ar-u$
 he(SUBJ)=TOP student=CONCEP-LOC-CONJ(ADJUNCT) exist-PRES
 'He exists, and it is as a student.'

which, in line with Section 2, can be schematically represented as the conjunctive proposition $E(A) \wedge IN(rE(A), B)$, is the **partial negation** in (44):

(44) $kare=wa$ $gakusei=de=wa$ $(*ara-)$ **-nai-0**
 he(SUBJ)=TOP student=CONCEP-LOC-CONJ(ADJUNCT)=SCOPE OF NEG exist NEG-PRES
 'He exists, but it is **not** as a student.'

rather than the total negation in (45):

(45) $kare=wa$ $gakusei=de$ $(*ara-)$ **-nai-0**
 he(SUBJ)=TOP student=CONCEP-LOC-CONJ(ADJUNCT) exist NEG-PRES
 'He does **not** exist as a student.'

Note that in (44) and (45), the stem of the existential verb *aru* is represented as (**ara-*), since in contemporary Standard Japanese the negative form of this verb is not the theoretically expected **ara-nai*, but obligatorily the suppletive adjective *nai*. The more natural sentence *kare=wa gakusei=de=wa nai* in (44) — or, more colloquially, *kare=wa gakusei=dya nai*, with the contracted form *=dya* derived from the sequence *=de=wa* — does **not negate the existence of *kare* itself**, but rather **negates only *gakusei=de* as the relevant domain of existence**. In this way, **the particle *=wa* in Japanese can mark the scope of negation**.

Schematically, the partial negation *A=wa B=de=wa nai* in (44) — underlyingly, **A=wa B=de=wa ara-nai* — can be represented as $E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$, whereas the total negation *A=wa B=de nai* in (45) — underlyingly, **A=wa B=de ara-nai* — can be represented as $\neg(E(A) \wedge \text{IN}(\text{rE}(A), B)) \equiv \neg E(A) \vee \neg \text{IN}(\text{rE}(A), B)$. Although this disjunction is compatible with $\neg E(A) \wedge \text{IN}(\text{rE}(A), B)$, $E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$ or $\neg E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$, the first case is inconsistent with the conditional $\text{IN}(\text{rE}(A), B) \Rightarrow E(A)$ — hence, the contrapositive $\neg E(A) \Rightarrow \neg \text{IN}(\text{rE}(A), B)$ — and the third case, $\neg E(A) \wedge \neg \text{IN}(\text{rE}(A), B) \equiv \neg E(A)$, is pragmatically infelicitous in topicalised constructions with *A=wa*, since *A=wa*, as a topicalised phrase, tends to presuppose the contextual or situational givenness — that is, the existence — of *A*, and since there are dedicated Japanese forms for existential negation such as *A=wa nai* / *i-nai* / *sonzai-si-nai*; in practice, therefore, the interpretation converges on $E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$. Hence, the former pattern in (44), which unambiguously denotes the partial negation $E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$, is judged to be the more natural one.

If *=de aru* were a single copula, then *A=wa B=de aru* would denote $\text{BE}(A, B)$, and its negation would take the form of plain predicate negation — namely, *A=wa B=de nai*, representable as $\neg \text{BE}(A, B)$, with no *=wa* following *=de*. **Empirically, however, *A=wa B=de nai* is perceived as infelicitous, whereas *A=wa B=de=wa nai* is preferred**. This spontaneous preference for the particle *=wa* after *=de* is unexpected under the simple-copula analysis, but follows naturally if the sentence is decomposed as $E(A) \wedge \text{IN}(\text{rE}(A), B)$, where the logically and pragmatically most plausible interpretation is the partial negation $E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$.

By contrast, the **default negation of the location-specifying-only** (46) and (47) = (22) and (23):

- (46) *sore=wa gakkō=ni ar-u*
 it(SUBJ)=TOP school=PHYSIC-LOC(ARGUMENT) be located-PRES
 'It is located in the school.'

- (47) kare=wa gakkō=ni i-ru
he(SUBJ)=TOP school= PHYSIC-LOC(ARGUMENT) be located-PRES
'He is located in the school.'

which can both be represented as $\text{IN}(A, B)$, is the **total negation** indicated in (48) and (49):

- (48) sore=wa gakkō=ni (*ara-) nai-0
it(SUBJ)=TOP school=PHYSIC-LOC(ARGUMENT) be located NEG-PRES
'It is **not** located in the school.'
- (49) kare=wa gakkō=ni i- nai-0
he(SUBJ)=TOP school= PHYSIC-LOC(ARGUMENT) be located NEG-PRES
'He is **not** located in the school.'

Unlike the existence-asserting and existence-domain-specifying conjunctive $E(A) \wedge \text{IN}(rE(A), B)$, the location-specifying-only $\text{IN}(A, B)$ is an atomic proposition and, naturally, its negation constitutes $\neg \text{IN}(A, B)$, which corresponds to (48) and (49).

On the other hand, when the PP *gakkō=ni* in (48) and (49) is marked by *=wa*, this particle functions contrastively, as in (50) and (51), respectively, where CONTR abbreviates contrastive:

- (50) sore=wa gakkō=ni=**wa** (*ara-) **nai-0**
it(SUBJ)=TOP school=PHYSIC-LOC(ARGUMENT)=CONTR be located NEG-PRES
'It is **not** located in the school **but somewhere else.**'
- (51) kare=wa gakkō=ni=**wa** i- **nai-0**
he(SUBJ)=TOP school= PHYSIC-LOC(ARGUMENT)=CONTR be located NEG-PRES
'He is **not** located in the school **but somewhere else.**'

In contrast to (48) and (49), which are representable as $\neg \text{IN}(A, B)$, (50) and (51) can be represented schematically as $\neg \text{IN}(A, B) \wedge \exists x (\text{IN}(A, x) \wedge x \neq B)$ — semantically, 'A is not located in B, and there exists a physical domain x distinct from B such that A is located in x'.

The difference in default negation pattern between the existence-asserting and existence-domain-specifying construction and the location-specifying-only one lies in the fact that the former, $E(A) \wedge IN(rE(A), B)$, is **conjunctive**, whereas the latter, $IN(A, B)$, is **atomic**. This ultimately derives from **the status of B as an adjunct in the former and as an argument in the latter**, since if B were an argument, $E(A) \wedge IN(rE(A), B)$ would correspond to something like $EXIST-MANNER(A, B)$, whose negation would be the total $\neg EM(A, B)$.

Finally, if *=de aru* were a single copular unit, partial negation with *=wa* would necessarily take

the form **kare=wa gakusei=wa de nai*, as in (52):

- (52) *kare=wa gakusei=**wa** de **nai**-0
 he(SUBJ)=TOP student(PRED)=CONTR COP NEG-PRES
 'He is **not** a student **but something else**.'

The fact that this is ungrammatical further supports the separability of *=de* and *aru* and the constituency of *gakusei=de*.

9. The differences between *A=wa B=de aru* and *A=ga B=de aru*

The following table (53) provides a preliminary overview of the propositional identity and the information-structural differences between sentences of the type *A=wa B=de aru* and those of the type *A=ga B=de aru*, without attempting a full formalisation, which lies beyond the scope of this paper:

(53)

Sentence	Propositional structure	Informational structure	
		Given	New
<i>A=wa B=de aru</i>	$E(A) \wedge \text{IN}(\text{rE}(A), B)$	$E(A)$	$\text{IN}(\text{rE}(A), B)$
<i>A=wa B=de=wa nai</i>	$E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$	$E(A)$	$\neg \text{IN}(\text{rE}(A), B)$
<i>A=wa B=de=wa naku</i> <i>C=wa B=de aru</i>	$E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$ $\wedge E(C) \wedge \text{IN}(\text{rE}(C), B)$	$E(A) \wedge E(C)$	$\neg \text{IN}(\text{rE}(A), B)$ $\wedge \text{IN}(\text{rE}(C), B)$
<i>A=ga B=de aru</i>	$E(A) \wedge \text{IN}(\text{rE}(A), B)$	$\exists x (E(x) \wedge \text{IN}(\text{rE}(x), B))$	$x = A$
<i>A=ga B=de=wa nai</i>	$E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$	$\exists x (E(x) \wedge \text{IN}(\text{rE}(x), B))$	$x \neq A$
<i>A=ga B=de=wa naku</i> <i>C=ga B=de aru</i>	$E(A) \wedge \neg \text{IN}(\text{rE}(A), B)$ $\wedge E(C) \wedge \text{IN}(\text{rE}(C), B)$	$\exists x (E(x) \wedge \text{IN}(\text{rE}(x), B))$	$(x \neq A) \wedge (x = C)$

Syntactically, the structure of *A=ga B=de aru* in (53) can be represented as in (54):

- (54) [_{TP} [_{NP} *A=ga_i*] [_{T'} [_{VP} [_{ConjP} *B=de*] [_{VP} [_{NP} *x_i*] [_V *ar-*]]] [_T *-ru*]]]

where *x* represents a trace, used here instead of the conventional *t*, and [_V *ar-*] has not been moved for reasons of visibility. In this structure, the VP [_{VP} [_{ConjP} *B=de*] [_{VP} [_{NP} *x_i*] [_V *ar-*]]] denotes the given information $\exists x (E(x) \wedge \text{IN}(\text{rE}(x), B))$, whereas the **untopicalised** — and thus capable of indicating a piece of new information — NP [_{NP} *A=ga_i*] denotes the new information $x = A$, through the anaphoric relation of the trace [_{NP} *x_i*] to its antecedent [_{NP} *A=ga_i*]. The constituents representing given and new information are enclosed within boxes and, furthermore, those representing new information are shaded. (The bracket notation adopted here is a

descriptive representation of hierarchical constituency, and the traditional category labels are used merely as a device for visualising the structure.)

On the other hand, the structure of $A=\textit{wa}$ $B=\textit{de aru}$ in (53) can be represented as in (55):

$$(55) [\text{TopP} [\boxed{A_i=\textit{wa}}] [\text{Top}' [\text{TP} [\text{NP } x_i] [\text{T}' [\text{VP} [\boxed{\text{ConjP } B=\textit{de}}] [\boxed{\text{VP} [\text{NP } x_i] [\text{V } \textit{ar-}]]}] [\text{T } \textit{-ru}]]] [\text{Top } \textit{-0}]]]$$

The category *Topic*, abbreviated as *Top* in (55), is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme *-0*. The element that moves to the specifier of *TopP* is assigned the particle $=\textit{wa}$ and functions to indicate the topic of the sentence. In this structure, the **topicalised** phrase $[A_i=\textit{wa}]$ — which typically indicates a piece of given information — and the VP $[\text{VP} [\text{NP } x_i] [\text{V } \textit{ar-}]]$ — where the verb $[\text{V } \textit{ar-}]$ 'to exist' provides another piece of given information, since the topicalised phrase $[A_i=\textit{wa}]$ generally presupposes the existence of A — together denote the given information $E(A)$. By contrast, the *ConjP* $[\text{ConjP } B=\textit{de}]$ denotes the new information $\text{IN}(\text{rE}(A), B)$, since a sentence conveying only given information would be informationally vacuous. Note that, since $\text{IN}(\text{rE}(x), B) \Rightarrow E(x)$, it follows that $\exists x (E(x) \wedge \text{IN}(\text{rE}(x), B)) \equiv \exists x \text{IN}(\text{rE}(x), B)$; nevertheless, the logically redundant $\exists x (E(x) \wedge \text{IN}(\text{rE}(x), B))$ represents a component within the informational structure that corresponds to the syntactic–semantic structure assumed in this paper.

10. The contracted form $=\textit{da}$

In spoken language, the contracted form $=\textit{da}$ is often used instead of $=\textit{de aru}$, which is more formal and literal, as in $\textit{kare}=\textit{wa}$ $\textit{gakusei}=\textit{da}$ rather than $\textit{kare}=\textit{wa}$ $\textit{gakusei}=\textit{de aru}$. It should be noted that this **contracted form $=\textit{da}$** does not represent a new copular category but is a **purely phonological reduction of the postpositional–conjunctive–verbal complex $=\textit{de aru}$** , as shown in (56):

$$(56) \begin{array}{lll} \textit{kare}=\textit{wa} & \textit{gakusei}=\textit{d} & \textit{a-0} \\ \text{he(SUBJ)}=\text{TOP} & \text{student}=\text{CONCEP-LOC-CONJ(ADJUNCT)} & \text{exist-PRES} \\ \text{'He exists, and his existence is located in the conceptual domain of "student".'} & & \end{array}$$

The historical change from $=\textit{de aru}$ to $=\textit{da}$, possibly via $=\textit{dea}$ ³, involves no syntactic or

³ Historically, the form $=\textit{dea}$, which yielded the modern $=\textit{da}$, can be attested in Azuchi–Momoyama-

semantic reanalysis, but a process of phonological contraction comparable to French *à le* → *au* 'to the' or Portuguese *de a* → *da* 'from the' or 'of the'. Just as there is no semantic necessity for *à* and *le* to fuse into *au*, or for *de* and *a* into *da* — a phenomenon comparable to the English *to the* or *from the* forming a single word — there is likewise no semantic motivation for *=de* and *aru* to form a single lexical item.

That said, however, the reduction of *aru* to *a* can be accounted for in pragmatic and informational terms. First, as already discussed in Sections 2 and 5, the crucial logical — though not syntactic — predicate in this construction is not the verbal predicate *ar-*, representable as $E(\quad)$, but the underlying predicate-like postposition *=ni*, representable as $IN(r(\quad), \quad)$, contained within the surface postposition–conjunctive complex *=de*. Second, since the topicalised phrase *A=wa* generally presupposes, at the discourse level, the existence of *A*, and since the occurrence of the constituent *B=de* entails the existence of *A* by virtue of the conditional $IN(rE(A), B) \Rightarrow E(A)$, the verb *aru* becomes informationally redundant when it once again asserts *A*'s existence. Third, in a sentence in which $IN(rE(A), B)$ is presented as new information, $E(A)$ is presupposed and thus treated as given, owing to the same conditional $IN(rE(A), B) \Rightarrow E(A)$. In this context, $E(A)$ is pragmatically and informationally less significant than the focus component $IN(rE(A), B)$. It may therefore be assumed that **this low pragmatic and informational salience is reflected in the phonological reduction of *aru* to *a***. This reduction can be analysed as follows: morphologically, the [–past] tense suffix takes the zero allomorph /-0/ rather than the canonical /-ru/ in *aru* /ar-ru/, yielding /ar-0/; phonologically, the word-final /-r/ is deleted at the phonetic level, giving rise to the surface form *a*.

The reduction of the underlying *=ni-te ar-0* to the surface *=da*, via *=de ar-0*, on the other hand, can be accounted for by the fixed order of these concatenated morphemes and its frequency. In existence-asserting and existence-domain-specifying constructions, the morphemes *=ni*, *-te* and *ar-* — which as a whole corresponds to the schematic expression $E(\quad) \wedge IN(rE(\quad), \quad)$ and in which *=ni*, *-te* and *ar-* represent $IN(r(\quad), \quad)$, $\wedge$ and $E(\quad)$, respectively — is invariably ordered in this way, except in scrambling constructions — or *kakimazekōbun* in Japanese linguistics — such as *B=de watasi / watasitati / wareware=wa ari-tai* '(lit.) as B, I / we / we want to exist'. **The very high frequency of this fixed order in actual utterances is assumed to have led to**

period materials: *bonnin yorimo giūzaini fuxōzuru cotodea* '(it) is a reason for imposing a heavier guilt (on him) than on an ordinary person' (Amakusa Edition of *Esopo no fabulas*, 1593, p.475, ll.10-11, in Text Data Sets for Research on the History of Japanese).

the phonological reduction of *=ni-te ar-0* to *=da*, via *=de ar-0*, just as in the French and Portuguese examples mentioned above, and likewise in the contractions *si-te=i-ru* → *si-te-ru* 'to be doing something' and *si-te=sima-u* → *si-t-ya-u* 'to end up doing sth.' or 'to do sth. inadvertently or unintentionally' in colloquial Japanese.

The three morphemes therefore remain distinct in their underlying structure: *=de* — underlyingly *=ni-te* — functions as a postposition–conjunction complex specifying adjunctively the conceptual domain in which existence holds, whereas the stem *ar-* of the verb *aru* serves as the existential predicate itself. This is precisely why the following highly colloquial dialogue in (57) is possible:

(57) *Kare=tte gakusei=da yone? — Iya, gakusei=tte yū=ka, gakusei=de=mo aru=tte kanzi kana.*

'He's a student, right?' — 'Well, if you can call him a student, I'd say he's *also* a student, kind of.'

In the dialogue (57), the interlocutor, recognising that the underlying structure of *gakusei=da* is *gakusei=de aru*, when asked to confirm that *kare*'s domain of existence can be identified with that of 'student', replies that his existence extends to other domains as well, expressed by the form *gakusei=de=mo aru*, representable as $E(\text{he}) \wedge \text{IN}(\text{rE}(\text{he}), \text{student}) \wedge \text{IN}(\text{rE}(\text{he}), \text{something else}) \dots$

Returning to Section 1, *kare=wa gakusei=de, kanozyo=wa syakaizin=da* 'he (is) a student, and she is a working adult' in (5) is thus, underlyingly, a compound sentence *kare=wa gakusei=de ari, kanozyo=wa syakaizin=de aru*, in which the first predicate *ari* constitutes presupposed given information and is therefore judged to be less informationally salient, and consequently deleted as recoverable, based on the structural parallelism between the two coordinate clauses, as indicated in (58):

(58)

(58)	<i>kare=wa</i>	<i>gakusei=de</i>	(ari)	<i>kanozyo=wa</i>	<i>syakaizin=d</i>	a-0
	he(SUBJ)=TOP	student=LOC(ADJC)	(exist)	she(SUBJ)=TOP	working adult=LOC(ADJC)	exist-PRES
	'He (exists) as a student, she exists as a working adult.'					

The sentence in (58) is an instance of zeugma, in which the inflected form *ari* of the existential verb *aru* is deleted owing to its low pragmatic and informational salience and to its contextual recoverability. Therefore, *=de* is not an inflected form of *=da*. Consequently, *=da* as a whole should not be regarded as a copula in the strict sense, but as a **phonologically reduced surface realisation of the underlying construction *=ni-te ar-0***, where *ar-* and *-0* represent the stem of

the existential verb *aru* and a phonologically null tense suffix specified as [–past], respectively.

11. Syntactic and logical-propositional differences between NPs with =*da*, nominal adjectives and verbal adjectives

Japanese **nominal adjectives** (NAs), also referred to as ***na*-adjectives** due to their attributive forms ending in *-na* — the so-called *rentaikei* in Japanese linguistics — as in *yukai-na hito* 'a pleasant person', have often been analysed as consisting of "so-called NA stems plus the so-called copula =*da*", in parallel with "NPs plus the so-called copula =*da*", as in *sore=wa yukai=da* 'it is pleasant'.

According to Makino (2025²: 22), the core of NAs lies in the NA stem in *-ni*, as in *yukai-ni* 'to be pleasant' and *suki-ni* 'to like'. The element *-ni* is not a postposition but the NA thematic suffix, which functions as an indispensable morphological and semantic constituent in the formation of the NA stem. These NA stems are lexical predicates that require arguments — the argument structure [theme] inherent in *yukai-ni* and [experiencer, theme] in *suki-ni*. However, outside small clauses functioning as complements of cognition or perception verbs, as in *watasi=wa [sore=o yukai-ni] omot-ta / kanzi-ta* 'I found / felt [it pleasant]', such NA stems cannot function as predicates in their bare *-ni* form.

Consequently, an NA stem forms a nominal-adjective phrase accompanied by PRO, as in [*PRO yukai-ni*], which serves as the complement of the adjunctive conjunction *-te*, yielding a ConjP [*PRO yukai-ni-te*]. This ConjP is then adjoined to the VP headed by the existential-verb stem *ar-* within a one-argument obligatory construction, as in [*sore_i=wa t_i ar-0*] 'it exists'. Only in this configuration can the NA stem function as the predicate of an independent sentence, as in [*sore_i=wa [PRO_i yukai-ni-te] t_i ar-0*], which surfaces as *sore=wa yukai-d-a* [*sore_i=wa [PRO_i yukai-d] t_i a-0*] 'it is pleasant' (literally, 'it exists and is pleasant'). On the other hand, the so-called copular construction *sore=wa gakkō=d-a* 'it is a school' can be underlyingly analysed as [*sore_i=wa [gakkō=ni-te] t_i ar-0*], which surfaces as [*sore_i=wa [gakkō=d] t_i a-0*].

Thus, the so-called copular sentences and the nominal-adjectival sentences share the syntactic property that they both consist of an existence-asserting, one-argument obligatory construction combined with either an existence-domain-specifying adjunctive ConjP or a state-specifying adjunctive ConjP, and can both be schematically represented as conjunctive propositions, as in

(59) and (60):

(59) *Sore=wa gakkō=d-a. [sore_i=wa [gakkō=ni-te] t_i ar-0]* $E(\text{sore}) \wedge \text{IN}(\text{rE}(\text{sore}), \text{gakkō})$

(60) *Sore=wa yukai-d-a. [sore_i=wa [PRO_i yukai-ni-te] t_i ar-0]* $E(\text{sore}) \wedge \text{PLEASANT}(\text{sore})$

The difference is that, since the PP [*gakkō=ni*] in (59) does not constitute a lexical predicate, it does not require any argument; therefore, the adjunctive construction [*gakkō=ni-te*] does not contain PRO, unlike the adjunct [*PRO yukai-ni-te*] in the nominal-adjectival construction in (60).

By contrast, **verbal adjectives (VAs)** — also referred to as ***i*-adjectives** due to their attributive forms ending in *-i*, as in *tanosi-i hito* 'a fun person' — form constructions in which they themselves function as the main predicate. Such constructions, as in *sore=wa tanosi-i* 'it is enjoyable', consist only of a state-specifying obligatory construction without any adjunct and can be schematically represented as atomic propositions, as in (61):

(61) *Sore=wa tanosi-i. [sore_i=wa t_i tanosi-i-0]* $\text{ENJOYABLE}(\text{sore})$

The syntactic and logical-propositional differences between (59) and (60), on the one hand, and (61), on the other, become manifest in their default negation patterns, as illustrated in (62) – (64):

(62) *Sore=wa gakkō=de=wa na-i.* $E(\text{sore}) \wedge \neg \text{IN}(\text{rE}(\text{sore}), \text{gakkō})$

(63) *Sore=wa yukai=de=wa na-i.* $E(\text{sore}) \wedge \neg \text{PLEASANT}(\text{sore})$

(64) *Sore=wa tanosi-ku na-i.* $\neg \text{ENJOYABLE}(\text{sore})$

(65) *Sore=wa tanosi-ku=wa na-i.* $\neg \text{ENJOYABLE}(\text{sore}) \wedge \text{SOME-OTHER-STATE}(\text{sore})$

The default negation patterns of the constructions in (59) and (60) — representable as the conjunctive propositions $E(\text{sore}) \wedge \text{IN}(\text{rE}(\text{sore}), \text{gakkō})$ and $E(\text{sore}) \wedge \text{PLEASANT}(\text{sore})$ — are partial, namely $E(\text{sore}) \wedge \neg \text{IN}(\text{rE}(\text{sore}), \text{gakkō})$ and $E(\text{sore}) \wedge \neg \text{PLEASANT}(\text{sore})$, as in (62) and (63), whereas that of the construction in (61) — representable as the atomic proposition $\text{ENJOYABLE}(\text{sore})$ — is total, namely $\neg \text{ENJOYABLE}(\text{sore})$, as in (64).

It follows that the stem allomorph *at-* of the verb *ar-u* occurring in the past form *tanosi-k-at-ta*, for example, does not instantiate the one-argument existential verb *ar-* 'to exist' used in both so-called copular and nominal-adjectival sentences. Rather, as stated in Makino (2005¹: 4), it is an allomorph of "the stem *ar-* from the light verb *ar-u*" inserted as a matter of formal necessity in order to concatenate the **verbal** past-tense suffix *-ta* with the **non-verbal** VA stem *tanosi-ku*, which is incompatible with the former, as in *tanosi-ku + -ta* → *tanosi-ku + ar- + -ta* → *tanosi-k-at-ta*. The default negation of this past form *tanosi-k-at-ta* is likewise total, namely *tanosi-ku na-k-a-ta*, rather than the partial negation *tanosi-ku=wa na-k-at-ta*. This demonstrates that *tanosi-ku* does not constitute an adjunct forming a separate proposition and thus provides further evidence that *ar-* in this construction is not a one-argument existential verb. Note finally that the partial negation of the construction in (61) gives rise to an implication such as $\wedge$ SOME-OTHER-STATE(sore).

12. Conclusion — *=da* as the predicate-like and conceptual-locative postposition *=ni* plus the adjunctive conjunction *-te* followed by the one-argument existential verb *aru*

In summary, **Japanese has no genuine copula — *da* is not *be***. The construction *A=wa B dearu* 'A exists, and its existence is located in the conceptual domain of B' consists of a one-argument, existence-asserting core construction [*A=wa aru*] 'A exists' and a existence-domain-specifying adjunct [*B=de*], meaning 'and (A's existence is located) in the conceptual domain of B' or, more practically, 'and (it is) as B'. It should therefore be syntactically represented as *A=wa B=de aru*. In this construction, the most crucial logical — though not syntactic — predicate is not the verbal stem *ar-* that asserts *A*'s existence, but rather the predicate-like postposition *=ni* within the postposition–conjunction complex *=de*, which locates *A*'s existence in the conceptual domain of *B*. Consequently, treating *dearu* or its contracted form *da* as a single copula is structurally problematic. The form *dearu* is more appropriately analysed as an existence-and-domain complex, consisting of the one-argument existential verb *aru* and the predicative and conceptual-locative postposition *=ni* plus the adjunctive conjunction *-te*, which surface as *=de* and conjunctively specify the domain in which the existence of the theme is realised. The more colloquial form *=da* is simply a phonologically contracted form corresponding to the more formal and literal *=de aru*.

Thus, through the above syntactic and semantic analysis, it has been possible to account, by means of a single, consistent principle — namely, the **assertion of existence** by $[NP_1=wa + ar-]$ and the **conjunctive specification of the domain of existence** by $[NP_2 + =de]$ — underlyingly $[NP_2 + =ni-te]$ — schematically representable as $E(NP_1) \wedge IN(rE(NP_1), NP_2)$, not reducible to $BE(NP_1, NP_2)$ — for:

- Morphosyntactic separability of *de* and *aru* in *dearu*
- Optional omission of *aru* in *dearu*
- Prosodic-word boundary between *de* and *aru*
- Syntactic and prosodic constituency of *de* and the NPs preceding it
- Grounds for analysing *=de* as an adjunct-marking postposition–conjunction complex that functions as an eventuality-domain specifier $\wedge (r(), NP)$
- Foregrounding of the lexical meaning 'to exist' of the verb *aru*
- Grounds on which the Japanese desiderative *=de ari-tai* is not equivalent to the English *to want to be*
- Diachronic split between $A B=ni ari \rightarrow A B=nite ari \rightarrow A=wa B=de aru$ and $A B=ni ari \rightarrow A=wa B=ni aru/iru$
- Grounds on which the default negation of $A=wa B=de aru$ is not the total $A=wa B=de nai$ but the partial $A=wa B=de=wa nai$
- Syntactic and information-structural differences between $A=wa B=de aru$ and $A=ga B=de aru$
- Underlying structure *=de ar-0* surfacing as *=da*
- Pragmatic and informational motivation for the phonological reduction $aru \rightarrow a$ in *=da*
- Commonalities and differences between the so-called copular constructions and the nominal-adjectival constructions

Finally, while the present study reinterprets *=da* and *=de aru* as a postpositional-conjunctive-verbal complex expressing $E(A) \wedge IN(rE(A), B)$ rather than a copular predicate denoting $BE(A,$

B), this proposal is not intended to invalidate previous research conducted within the copular framework. Much of the empirical and descriptive work on so-called copular sentences remains valuable and can, in fact, be accommodated within the present account as analyses of the $\text{IN}(\text{rE}(\text{A}), \text{B})$ component — that is, as investigations into how the domain of existence is structured. In this respect, the present analysis may be regarded not as a rejection of the copular tradition, but as an attempt to extend it by situating its insights within a broader model of Japanese predication structure. Furthermore, the interaction between the existence-asserting component $\text{E}(\text{A})$ and the domain-specifying component $\text{IN}(\text{rE}(\text{A}), \text{B})$, which has received relatively little explicit attention in previous research, may offer a possible perspective for future studies on how these two semantic components relate to one another in information structure, negation and discourse.

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